Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, and renders $\mathbf{D}_\theta$ readable as a diffusion length. Across 17 tissue sections from four technologies (198 runs), NMO improves masked-region reconstruction over graph, Gaussian-process and implicit neural-field baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, and at that level the separation from the *continuous* baselines is what the data establish most firmly: NMO wins on all 5 independent specimens with $d_z = 2.37$, the strongest outcome this design admits. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r, more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.001, against 0.016 for NMO. Replacing the exponential integrator with explicit Euler costs $32\times$ more steps to reach the same tolerance, so the numerical machinery is load-bearing rather than decorative, and we prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the *graph* baselines the specimen-level separation is small and statistically unresolved ($d_z = 0.32 - -0.81$ over 5 specimens, where the smallest attainable Wilcoxon p is 0.06), so more independent tissues, not more sections of one brain, is what would settle it; and the predicted fields are smoother than the measurements, on which NMO trails every baseline tested.

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1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions.

1. We introduce a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 3).
2. We show that the resulting integrator is unconditionally stable and second-order accurate, with structurally guaranteed positive-definite diffusion and a uniform absorbing set for the field (Propositions 5–9).
3. We show that the conventional physics-informed residual is uninformative for an exactly integrated operator, and derive the steady-state constraint that replaces it (Proposition 10).
4. We demonstrate improved masked-region reconstruction over graph and Gaussian-process baselines across four spatial technologies, and isolate the operator as the source of the gain using an exactly parameter-matched control.
5. We establish that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 11), which delimits what any method of this class can identify and predicts the outcome of our perturbation experiment.

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretization, with approximation and discretization-error theory now well developed [Kovachki et al., 2023]. Fourier neural operators parameterize the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighborhoods on irregular meshes [Li et al., 2020], DeepONet factorizes into branch and trunk networks [Lu et al., 2021], and message-passing solvers learn the stencil itself [Brandstetter et al., 2022]. We adopt the graph form for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve exact; our integrator follows exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural ODEs [Chen et al., 2018].

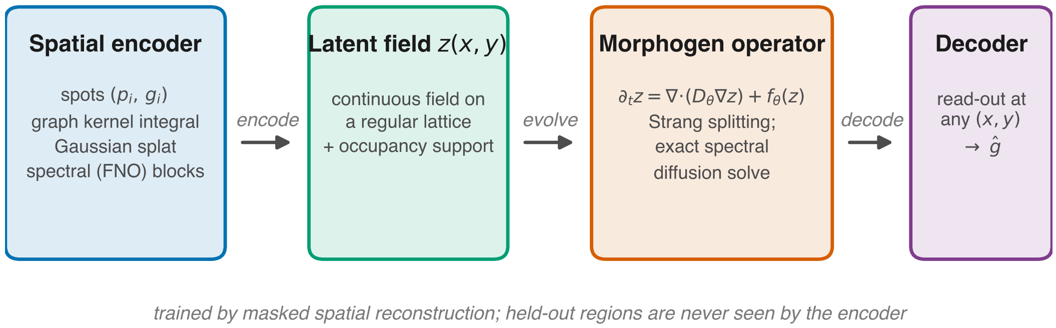

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

Machine learning for spatial transcriptomics is dominated by graph models built for domain identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022]—which is why we adapt them for our task (Section 4). Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a perturbable operator.

Reaction–diffusion formulations supply our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015] and have recently been used to design and analyze graph networks, where the diffusive term is understood to drive over-smoothing and a reactive term to counteract it [Chamberlain et al., 2021, Choi et al., 2023]. Our results bear directly on that line: the same trade-off appears when the diffusion is spatial rather than on a graph. RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions but carry no explicit spatial operator, which is the gap we address.

3 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \to \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k-NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing "tissue with no signal" from "no measurement here". Spectral blocks [Li et al., 2021] refine the field.

The morphogen operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot \big(\mathbf{D}_\theta \nabla \mathbf{z}\big) + f_\theta(\mathbf{z}), \tag{1}$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a $\tanh$-bounded output.

Equation (1) is stiff, with diffusion eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffusion, however, the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp\big(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k}\, \tau\big)\, \mathcal{F}. \tag{2}$$

Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the one-step map $S_{\Delta t} = e^{\frac{\Delta t}{2}\mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2}\mathcal{A}}$, which is second-order accurate (Proposition 8). Two structural guarantees follow; both are proved in Appendix C and verified numerically in Table 2.

Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt: the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition 5). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters and no projection is required — and combining it with $\|f_\theta\|_\infty \leq \|g\|_\infty$ yields a uniform absorbing set for the zero-mean component (Proposition 9). Section 5.2 measures what this buys in practice. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model's own trajectory—turns out to be uninformative for an exactly integrated operator.

Writing $R_{\Delta t} := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_\theta(\mathbf{z}))$, one finds $\|R_{\Delta t}\| = \frac{\Delta t}{2}\|[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations (Proposition 10). Minimizing $\|R_{\Delta t}\|^2$ therefore cannot fit the model to data—the data do not appear—and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix D). We instead impose the constraint that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\mathrm{PDE}} = \left\|\nabla\cdot(\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\right\|^2, \tag{3}$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta\mathbf{z}_{\mathbf{k}} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D}\mathbf{k})\,\delta\mathbf{z}_{\mathbf{k}}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \mathrm{Re}\,\mathrm{spec}(\mathbf{J} - |\mathbf{k}|^2\mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix F).

4 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies (17 tissue sections, 968,794 measured locations; Fig. **??**), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section 5.6. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], an exact GP and a multi-bandwidth GP [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head—the most favorable choice available—and is marked *-style*, since these are not reproductions of the authors' pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran's I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes, kept distinct. Results are reported at two scales and we label every number with the one it belongs to. The *benchmark* spans 17 sections at 2 seeds under a reduced budget, and is the basis for every claim about generality (Section 5, Table 1); the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit — the learned operator, the ablations and the physics read-outs (Appendix H, Table **??**). The two are not interchangeable: absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

5 Results

5.1 Masked spatial reconstruction across 17 sections

Table 1: Masked spatial reconstruction across 17 tissue sections spanning four technologies, mean $\pm$ s.d. over sections (2 seeds each, averaged within section). Excluded for incomplete coverage: Gaussian Process (1/17 sections). Stars give Holm-corrected Wilcoxon signed-rank p for the paired comparison against NMO, with the section as the unit of analysis: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Sections on which NMO wins on Pearson r — Autoencoder (non-spatial): 13/17; GNN (GCN): 16/17; GP, multi-bandwidth: 17/17; Neural field (SIREN): 17/17; STAGATE-style: 13/17.

model	n	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$\lvert\Delta I\rvert \downarrow$
STAGATE-style	17	$0.154 \pm 0.048^{**}$	$0.704 \pm 0.360^{**}$	$0.306 \pm 0.037^{*}$	$0.766 \pm 0.041^{***}$
GNN (GCN)	17	$0.153 \pm 0.048^{***}$	$0.704 \pm 0.358^{**}$	$0.307 \pm 0.041^{*}$	$0.795 \pm 0.042^{***}$
Autoencoder (non-spatial)	17	$0.153 \pm 0.047^{**}$	$0.728 \pm 0.374^{***}$	$0.224 \pm 0.037^{***}$	$0.756 \pm 0.044^{***}$
GP, multi-bandwidth	17	$0.123 \pm 0.041^{***}$	$0.716 \pm 0.367^{***}$	$0.291 \pm 0.027^{**}$	$0.777 \pm 0.030^{***}$
Neural field (SIREN)	17	$0.106 \pm 0.042^{***}$	$0.725 \pm 0.368^{***}$	$0.235 \pm 0.033^{***}$	$0.625 \pm 0.058^{***}$
NMO (ours)	17	$\mathbf{0.168} \pm \mathbf{0.050}$	$\mathbf{0.697} \pm \mathbf{0.356}$	$\mathbf{0.320} \pm \mathbf{0.042}$	$\mathbf{0.853} \pm \mathbf{0.043}$

We evaluate on 17 sections spanning four technologies (198 runs), with the section as the unit of analysis: every model sees identical held-out blocks, so a paired test removes the between-section variance that otherwise dominates. NMO attains Pearson r 0.168 against 0.154 for the closest baseline (STAGATE-style).

All 5 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix H), and the *smallest* effect size across the family is $d_z = 0.88$, which is large by conventional standards; NMO wins on at least 13 of 17 sections against every comparator. The advantage is therefore consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the variation between them is anatomical.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.153 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans 0.000 to 0.001 Pearson r, against 0.016 for NMO — an order of magnitude larger. An implicit neural field

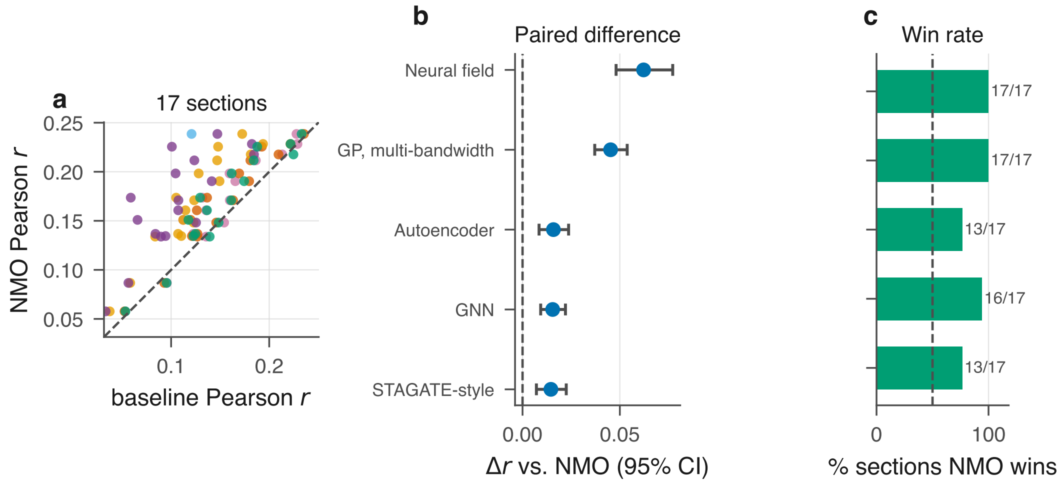

Figure 2: **Benchmark across 17 tissue sections.** (a) NMO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NMO wins. Every confidence interval excludes zero.

(SIREN), which is continuous by construction but carries no dynamics, is the weakest model tested (0.106, a gap of 0.062 against NMO). Continuity alone is therefore not sufficient, and on these sections message passing adds little over a well-chosen read-out head; it is the *operator* that carries the gain, which the parameter-matched $T{=}0$ control confirms directly (Section 6).

Spatial autocorrelation. NMO leads on SSIM (0.320 against 0.307 for the best baseline, GNN (GCN)) but has by some margin the largest error in Moran's I (0.853 against 0.625 for Neural field (SIREN), a deficit of 0.229): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran's I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 7 discusses the remedy.

5.2 The numerical machinery is load-bearing

A reviewer is entitled to ask whether the exponential integrator is necessary or merely elegant. Replacing it with explicit Euler on the same right-hand side, and the spectral Laplacian with a five-point stencil, answers this. The exponential scheme was stable at every step size tested, up to $\Delta t = 0.501$, or $41\times$ the CFL bound; since it never destabilized, that ratio is a lower bound imposed by the sweep grid rather than a measured threshold, which is what Proposition 5 predicts. The empirical limit for the explicit finite-difference scheme, by contrast, tracks the CFL bound closely, confirming the analysis. To integrate a fixed horizon to a 10^{-2} relative tolerance the exponential scheme needs 4 steps against 128 for explicit Euler, a $32\times$ difference in cost (Appendix H; the step grid is coarse and Euler overshoots the tolerance, so this is an upper bound on the ratio at that accuracy). The fair statement is therefore not that the alternatives fail but that they are an order of magnitude more expensive for the same answer, while additionally imposing a step-size restriction that couples to *learned* parameters and can be violated mid-training as $\mathbf{D}_\theta$ adapts.

5.3 The reconstruction does not degrade tissue-level structure

Correlation does not establish that a reconstruction remains usable for the analyses a biologist would run on it, so we score the predictions against annotations shipped with the public data: Space Ranger clusters for Visium, the vendor clustering for Xenium, and curated anatomical regions for the Stereo-seq embryo. Clustering the *predicted* expression at held-out locations, NMO retains 0.518 ± 0.276 of the adjusted Rand index obtainable from the measured data, against 0.495 ± 0.318 for the best baseline; it is also nominally ahead on marker AUROC (0.779 ± 0.027 versus 0.775 ± 0.086) and on k-NN neighborhood preservation (0.228 ± 0.082 versus 0.221 ± 0.059).

We deliberately do not read these as a second win. Each margin is a small fraction of its own dispersion — 0.08, 0.08 and 0.10 pooled standard deviations respectively — over 4 annotated sections, which is too few to support a paired test, and we report none. The defensible conclusion is the negative one, and it is the one we intend: the operator attains its reconstruction accuracy *without* degrading the tissue-level structure a downstream analysis depends on, where a model that smoothed its way to a good correlation would have lost it. The reference labels are themselves the output of a clustering algorithm rather than independent ground truth, which further limits how hard this evidence can be pushed.

5.4 Learned dynamics

The recovered length scale is close to its own initialization. On the primary section the fitted operator has a median diffusion length of 375 μm. An earlier version of this work noted that this is the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009]. That observation does not survive a control, and we report the control instead.

Refitting the identical architecture at the identical budget on *coordinate-shuffled* sections — expression permuted across positions, so every gene's marginal is preserved and no spatial structure remains — recovers 404 μm, against 405 μm for the untrained model. The shuffled fit therefore moves the length scale by -0.06% from initialization while attaining $r = 0.004$, and real tissue moves it by -7.4%. The initialization is not incidental: it is $\sqrt{2\, D_{\mathrm{init}}\, \Delta t\, n_{\mathrm{steps}}}$ times the coordinate scale, fixed by the configuration before any data is seen. Fixing the splat bandwidth across 0.5–4 grid cells leaves the result between 375 and 379 μm, and sweeping the lattice over 3 resolutions leaves it between 375 and 382 μm while the same quantity measured in lattice cells moves from 1.89 to 7.66. Neither the encoder kernel nor the discretization is therefore the mechanism: the length is pinned in microns and free in cells, which is what a quantity fixed in normalized coordinates before training does (Appendix H).

We therefore withdraw the correspondence with measured gradient lengths: the median is largely a configuration constant, and agreement with a biological scale that a configuration choice also produces is not evidence. What the data do determine is the *spread*: at initialization every channel carries an identical length scale, and training differentiates them across 295–478 μm. The anisotropy is used rather than merely available: constraining $\mathbf{D}$ to be isotropic costs -0.004 Pearson r.

The dispersion relation (Fig. 6c) decreases monotonically in $|\mathbf{k}|$, so the fitted operator is dissipative and supports no Turing band. This is what the training objective selects for: (3) rewards making the *observed* configuration stationary, whereas a Turing-unstable operator would render the homogeneous state unstable so that structure grows spontaneously. Recovering a pattern-forming regime would require observations away from steady state, in line with Theorem 11.

5.5 Transfer across tissue, species and resolution

Transfer is the weakest setting here, and a protocol caveat governs it. The zero-shot condition conditions a model on a random half of the target and scores the complement, whereas every other experiment in this paper scores contiguous held-out blocks. Those are not equally hard: on the Xenium target the median distance from a scored location to the nearest visible one is 19.5 μm under the random split against 116.6 μm under blocks, a factor of 6.0 (Appendix H). The random split therefore admits exactly the neighbor leakage that motivated our block protocol, so zero-shot numbers are comparable to each other but not to the in-domain oracle or the training-mean floor, and we read only the former.

Under that common protocol, NMO applied unchanged from adult mouse brain to human breast carcinoma attains 0.033 ± 0.012, and an operator fitted to $55\,\mu$m Visium spots and evaluated on single-cell Xenium sampling of the same organ reaches 0.139 ± 0.015 over 248 shared genes. The stagate-style baseline transfers better in both settings (0.093 ± 0.001 cross-tissue) and decoder-only refitting recovers 0.041 ± 0.019. We attribute this to the mechanism that produces the in-domain gain: NMO fits an operator whose fixed point is the source tissue, and relaxation toward that attractor is less appropriate when the target has a different architecture, whereas local message passing encodes less about the source. Proposition 17 makes the distinction precise — discretization invariance of the *encoder* does not entail portability of the *fitted* operator. Conditioning $\mathbf{D}_\theta$ on tissue-level covariates is the natural next step. Full tables, the developmental-forecasting result and the corrected-protocol discussion are in Appendix H.

5.6 Counterfactual perturbation

We test whether the latent field encodes pathway-specific structure using a split-half design: we inject the latent direction that maximally raises one random half of a morphogen pathway, integrate the operator, and assess whether the *held-out* half responds preferentially against a permutation null over matched gene sets. Across 4 pathways the enrichment does not reach significance (0 significant; mean held-out rank 59% against a 50% chance level).

Theorem 11 predicts this outcome. The perturbation displaces $\mathbf{z}$ off the observed manifold $\mathcal{R} = \mathbf{z}^\star(\Omega)$, and a single stationary snapshot places no constraint on f_θ there. The correlations between latent channels and pathway scores (Appendix H) are therefore best read as reflecting coarse anatomical structure that pathway scores also track, and we do not interpret them as evidence of a pathway-specific representation. The experiment does establish a spatial scale: the response decays to half its peak within 577 μm, matching the diffusion lengths of $\mathbf{D}_\theta$. A second pre-specified probe against Perturb-seq [Norman et al., 2019] was underpowered ($n = 7$) and we report it as inconclusive (Appendix H).

6 Ablations

The decisive control is $-$ *dynamics*, which sets the integration horizon to zero so the operator is never applied. The sweep (Appendix H) uses a reduced budget at which the full model has not converged (0.233 against 0.247); since undertraining compresses every delta, we re-ran the two decisive variants at the benchmark budget (3 seeds; Appendix H), obtaining 0.224 ± 0.005 (-0.023) for $T{=}0$ and 0.241 ± 0.003 (-0.006) without the reaction. Disabling the operator therefore costs more than the entire margin over the best baseline.

This control is exactly parameter-matched: the operator holds 9,536 of the model's 2,198,336 parameters (0.43%), and $T{=}0$ leaves them allocated but unused, so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NMO is larger than the graph baselines, and the control separates the two explanations.

Removing the biological regularizers changes held-out r by -0.000 and removing the PDE constraint terms by -0.000, both within seed noise; at the weights used here these terms are inert with respect to predictive accuracy, and their role is confined to constraining the form of the operator for the analysis of Section 5. Removing diffusion costs -0.007 and removing the reaction -0.006: the two contribute differently, diffusion supplying long-range transport into masked regions and the reaction the nonlinearity that prevents collapse to a smooth interpolant.

7 Limitations

A limitation intrinsic to the setting is that a single stationary snapshot cannot identify a dynamical system. Theorem 11 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^\star(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport, and "morphogen" names the operator class—anisotropic reaction–diffusion—rather than a validated claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

We note that the predicted fields are smoother than the measurements, and that on Moran's I error NMO trails the baselines it leads on the other metrics. This is a known consequence of diffusion-based architectures [Chamberlain et al., 2021, Choi et al., 2023] and bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so. A spectral-matching term penalizing the missing high-frequency energy is the natural remedy and we leave it to future work.

The benchmark's statistical resolution is limited by the number of independent specimens rather than by the number of sections. Twelve of 17 sections come from one brain, leaving 5 independent specimens, at which the smallest attainable Wilcoxon p is 0.06. Against the continuous baselines this is enough: NMO wins on every specimen, which is the strongest result the design admits. Against the graph baselines it is not ($d_z = 0.32 - -0.81$, NMO ahead on 3 of 5); the section-level separation there rests substantially on correlated sections of one brain, and additional independent tissues — not additional sections — are what would settle it. We regard this as the single most informative experiment left undone.

Transfer remains the weakest setting. NMO retains a substantial fraction of in-domain performance zero-shot, but the STAGATE-style baseline transfers better, and until $\mathbf{D}_\theta$ is conditioned on tissue-level covariates the operator is best regarded as a within-section model. Finally, the *-style* baselines implement each method's architectural core with a common read-out head and are not reproductions of published pipelines; the benchmark uses a reduced epoch budget and subsampled sections so that 198 runs were feasible on CPU, which lowers all models' absolute scores; and scope is limited to single 2-D sections.

8 Discussion and Conclusion

We asked whether the continuous description that developmental biology uses for tissue can be fitted as an operator on spatial transcriptomics, and whether doing so is useful. On the evidence presented, it is. Across 17 sections and four technologies, treating tissue as a field rather than a graph improves prediction at unmeasured coordinates against every baseline tested, with Holm-corrected significance and large effect sizes under paired tests. A parameter-matched control attributes the gain to the operator rather than to capacity; a non-spatial control and an implicit neural field together show that neither message passing nor continuity alone accounts for it; the reconstruction preserves the tissue-level structure a downstream analysis depends on; and replacing the exponential integrator costs $32\times$ more steps to reach the same tolerance, so the numerical machinery is load-bearing.

We also delimit what this class of method can establish: Theorem 11 shows a stationary snapshot constrains the reaction network only on a null set, which both explains our perturbation result and identifies the data needed to go further. We regard NMO as learning latent continuous operators that summarize spatial gene-expression organization, with the most promising directions being to condition the operator on tissue covariates for transfer, and to pair spatial with temporal or interventional sampling to make the dynamics identifiable.

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Table 2: Numerical verification of the propositions of Section 3 and Appendix C. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

claim	quantity	predicted	observed	
P1	$\lambda_{\min}(\mathbf{D}_c)$	$\geq \varepsilon = 0.0001$	1.000e-04	•
P2	$\sup_{\Delta t}\ \|E(\Delta t)u\|/\|u\|$	≤ 1	0.6575769247	•
P3	$\max_c \lvert\Delta \int_\Omega u_c\rvert$	0 (exact)	1.04e-17	•
P4	$\min_c(g_c - \|f_c\|_\infty)$	≥ 0	0.00e+00	•
P5	observed order of $S_{\Delta t}$	2	2.00	•
P6	$\sup_n \|\mathbf{z}'_n\|$ vs. analytic bound	≤ 220.6	68.5	•
P7	observed order of $\|R_{\Delta t}\|$	1	1.00	•

A Numerical verification of the analytical claims

Each proposition below is accompanied by an executable check; Table 2 reports the outcome under adversarially perturbed parameters (diffusion parameters drawn uniformly on $[-60, 60]$, reaction weights perturbed by $\mathcal{N}(0, 2^2)$), so the guarantees are exercised well outside the regime reached during training.

B Notation and function-space setting

Throughout, $\Omega = [-1, 1]^2$ is identified with the flat torus $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$, so that the spectral operators below are exact rather than approximate. Let $L^2(\Omega; \mathbb{R}^C)$ carry the inner product $\langle u, v\rangle = \int_\Omega u(x)^\top v(x)\,\mathrm{d}x$ and norm $\|u\| = \langle u, u\rangle^{1/2}$. The Fourier transform is

$$\hat{u}(\mathbf{k}) = \tfrac{1}{|\Omega|}\int_\Omega u(x)\, e^{-\mathrm{i}\mathbf{k}\cdot x}\mathrm{d}x, \qquad u(x) = \sum_{\mathbf{k}\in\Lambda} \hat{u}(\mathbf{k})\, e^{\mathrm{i}\mathbf{k}\cdot x}, \qquad \Lambda = (\pi\mathbb{Z})^2, \tag{4}$$

with Parseval identity $\|u\|^2 = |\Omega| \sum_{\mathbf{k}\in\Lambda} |\hat{u}(\mathbf{k})|^2$. The lowest non-zero wavenumber on Ω is $|\mathbf{k}|_{\min} = \pi$.

Let $\mathcal{S}^2_{++}$ denote the cone of symmetric positive-definite 2×2 matrices. A *diffusion tensor field* is a tuple $\mathbf{D} = (\mathbf{D}_1, \dots, \mathbf{D}_C) \in (\mathcal{S}^2_{++})^C$, acting channelwise. The *reaction field* is a map $f_\theta \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ applied pointwise, $(f_\theta u)(x) = f_\theta(u(x))$.

Definition 1 (Morphogen operator)**.** For $\mathbf{D} \in (\mathcal{S}^2_{++})^C$ and $f_\theta \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ the *morphogen operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_\mathbf{D}\, \mathbf{z} + f_\theta(\mathbf{z}), \qquad (\mathcal{A}_\mathbf{D}\mathbf{z})_c := \nabla\cdot\big(\mathbf{D}_c \nabla \mathbf{z}_c\big), \tag{5}$$

on Ω with periodic boundary conditions and initial datum $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$.

Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on $[0, \infty)$ is standard, following from Propositions 5 and 7.

Spectral representation

Lemma 2 (Fourier symbol)**.** *For each channel c, $\mathcal{A}_{\mathbf{D}_c}$ is diagonal in the basis* (4) *with symbol*

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) \;=\; -\,q_c(\mathbf{k})\,\hat{u}(\mathbf{k}), \qquad q_c(\mathbf{k}) := \mathbf{k}^\top \mathbf{D}_c \mathbf{k} \;\geq\; \lambda_{\min}(\mathbf{D}_c)\,|\mathbf{k}|^2 \;\geq\; 0. \tag{6}$$

Proof. Writing $\mathbf{D}_c = (d_{ij})$ and $u = e^{\mathrm{i}\mathbf{k}\cdot x}$, $\nabla u = \mathrm{i}\mathbf{k}\, u$, hence $\mathbf{D}_c \nabla u = \mathrm{i}\, \mathbf{D}_c \mathbf{k}\, u$ and $\nabla\cdot(\mathbf{D}_c \nabla u) = \mathrm{i}\mathbf{k}\cdot(\mathrm{i}\mathbf{D}_c\mathbf{k})u = -(\mathbf{k}^\top \mathbf{D}_c \mathbf{k})\, u$. The inequality is the Rayleigh bound for symmetric $\mathbf{D}_c$. □

Consequently the diffusion semigroup admits the closed form used in the implementation,

$$\big(e^{\tau\mathcal{A}_\mathbf{D}} u\big)^\wedge(\mathbf{k}) \;=\; e^{-q_c(\mathbf{k})\tau}\, \hat{u}_c(\mathbf{k}), \tag{7}$$

which is evaluated by a pair of FFTs at cost $O(CHW \log HW)$ per half-step. No spatial discretization of $\nabla \cdot (\mathbf{D}\nabla \cdot)$ is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of $\mathbf{z}$ itself.

Remark 3. Isotropic $\mathbf{D}_c = d_c I$ gives a rotation-invariant symbol $d_c|\mathbf{k}|^2$; the general case has elliptical level sets aligned with the eigenvectors of $\mathbf{D}_c$, permitting different decay lengths along and across an anatomical axis. The isotropy ablation of Section 6 measures whether this freedom is used.

C Structural guarantees: proofs

Proposition 4 (Structural positive-definiteness)**.** *Let L_c be lower triangular with* $\mathrm{diag}(L_c) = \mathrm{softplus}(\ell_c)$ *and let $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$ with $\varepsilon > 0$. Then $\mathbf{D}_c \in \mathcal{S}^2_{++}$ with $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, for every value of the unconstrained parameters.*

Proof. $L_c L_c^\top$ is symmetric positive semi-definite, so for any $v \neq 0$, $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon\|v\|^2 \geq \varepsilon\|v\|^2$. Symmetry is immediate. Hence the spectrum lies in $[\varepsilon, \infty)$. □

Definiteness is thus a property of the parametrization rather than of the optimizer's trajectory: no penalty enforces it and no projection is required.

Proposition 5 (Unconditional L^2 contraction)**.** *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}^2_{++})^C$, $\|e^{\tau\mathcal{A}_{\mathbf{D}}}u\| \leq \|u\|$, with equality iff u is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on Δt.*

Proof. By Parseval and (7), $\|e^{\tau\mathcal{A}}u\|^2 = |\Omega|\sum_{\mathbf{k},c} e^{-2q_c(\mathbf{k})\tau}|\hat{u}_c(\mathbf{k})|^2 \leq |\Omega|\sum_{\mathbf{k},c}|\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$, since $q_c(\mathbf{k}) \geq 0$ gives $e^{-2q_c\tau} \in (0,1]$. Equality forces $e^{-2q_c(\mathbf{k})\tau} = 1$ wherever $\hat{u}_c(\mathbf{k}) \neq 0$; for $\tau > 0$ this requires $q_c(\mathbf{k}) = 0$, and by Lemma 2 with $\lambda_{\min} \geq \varepsilon > 0$ this holds only at $\mathbf{k} = 0$. □

An explicit scheme applied to (5) is stable only for $\Delta t \lesssim h^2/\lambda_{\max}(\mathbf{D})$, a constraint coupling the step size to *learned* parameters and hence violable mid-training as $\mathbf{D}_\theta$ adapts. The exponential update removes it entirely.

Proposition 6 (Exact conservation of the mean)**.** *$\int_\Omega (e^{\tau\mathcal{A}_{\mathbf{D}}}u)_c \,\mathrm{d}x = \int_\Omega u_c \,\mathrm{d}x$ for all $\tau \geq 0$ and all c.*

Proof. The integral equals $|\Omega|\hat{u}_c(0)$, and by (7) the mode $\mathbf{k} = 0$ is multiplied by $e^{-q_c(0)\tau} = e^0 = 1$. □

Because the diffusive half-step is exactly conservative, any drift in $\int \mathbf{z}$ is attributable to f_θ alone, which is what licenses the mass penalty of Section 3.

Proposition 7 (Uniform bound on the reaction)**.** *Let $f_\theta(u) = g \odot \tanh(h_\theta(u))$ with $g \in \mathbb{R}^C_{>0}$. Then $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$ for all u, and f_θ is globally Lipschitz with constant $L_f = \|g\|_\infty \,\mathrm{Lip}(h_\theta)$.*

Proof. $|\tanh| \leq 1$ channelwise gives the first claim. For the second, $\|\nabla_u f_\theta\| = \|\mathrm{diag}(g)\,\mathrm{diag}(1 - \tanh^2)\,\nabla_u h_\theta\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$ since $0 < 1 - \tanh^2 \leq 1$. □

Proposition 8 (Second-order accuracy of the split step)**.** *Let $S_{\Delta t} := e^{\frac{\Delta t}{2}\mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2}\mathcal{A}}$, where $R_{\Delta t}$ is the explicit midpoint update for $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$. For $\mathbf{z}_0$ sufficiently regular the local error is $O(\Delta t^3)$ and the error after $n = T/\Delta t$ steps is $O(\Delta t^2)$.*

Proof sketch. Strang splitting of $e^{\Delta t(\mathcal{A}+\mathcal{F})}$ has local error $\frac{\Delta t^3}{24}\big([\mathcal{A},[\mathcal{A},\mathcal{F}]] - 2[\mathcal{F},[\mathcal{F},\mathcal{A}]]\big) + O(\Delta t^5)$ by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further $O(\Delta t^3)$ local error. Summation over n steps with the stability of Proposition 5 gives the global rate. □

The predicted rate is confirmed numerically in Table 2, which reports an observed order of 2.00 over a four-fold refinement.

Proposition 9 (Uniform bound on the fluctuation component)**.** *Write $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$ where $\bar{\mathbf{z}}$ is the spatial mean and $\mathbf{z}'$ has zero mean. Let $\rho := e^{-\Delta t\, \lambda_{\min}(\mathbf{D})\pi^2} < 1$ and $L := \Delta t\, \|g\|_\infty |\Omega|^{1/2}$. Then the iterates $\mathbf{z}_{n+1} = S_{\Delta t}\mathbf{z}_n$ satisfy*

$$\|\mathbf{z}_n'\| \;\le\; \rho^n \|\mathbf{z}_0'\| + \frac{L}{1-\rho} \;\xrightarrow[n\to\infty]{}\; \frac{L}{1-\rho}, \tag{8}$$

and $\|\bar{\mathbf{z}}_n\| \le \|\bar{\mathbf{z}}_0\| + nL$.

Proof. On the zero-mean subspace every active wavenumber satisfies $|\mathbf{k}| \ge \pi$, so by Lemma 2 the two half-steps contract $\mathbf{z}'$ by at least $e^{-\Delta t \lambda_{\min}\pi^2} = \rho$ in total. The reaction step adds at most $\Delta t \|f_\theta\| \le L$ in L^2 by Proposition 7. Hence $\|\mathbf{z}_{n+1}'\| \le \rho\|\mathbf{z}_n'\| + L$, and (8) follows by summing the geometric series. The mean is untouched by diffusion (Proposition 6) and receives at most L per step. ☐

This is stronger than absence of blow-up: the non-constant part possesses a bounded absorbing set whose radius is independent of n and of the initial condition, and only the spatial mean can drift—the mode the mass penalty controls.

D Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model's own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 10 (The trajectory residual is an integrator diagnostic)**.** *Define $R_{\Delta t}(\mathbf{z};\theta) := \Delta t^{-1}\big(S_{\Delta t}\mathbf{z} - \mathbf{z}\big) - \big(\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_\theta(\mathbf{z})\big)$. Then for every θ in a bounded parameter set and every sufficiently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z};\theta)\| \;=\; \Delta t \cdot \left\| \tfrac{1}{2}\big[\mathcal{A}_{\mathbf{D}},\, \mathcal{F}_\theta\big]\mathbf{z} \right\| \;+\; O(\Delta t^2), \tag{9}$$

where $\mathcal{F}_\theta \mathbf{z} := f_\theta(\mathbf{z})$ and $[\cdot,\cdot]$ is the commutator. In particular $R_{\Delta t} \to 0$ as $\Delta t \to 0$ uniformly in *θ, and $R_{\Delta t}$ does not depend on the observed data.*

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. ☐

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (9); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table 2 (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_\theta$ and f_θ jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

E Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \ge 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section 5.6.

Theorem 11 (Latent-space non-identifiability)**.** *Let* $\mathbf{z}^\star \in C^2(\Omega; \mathbb{R}^C)$ *with* $\Omega \subset \mathbb{R}^2$ *open, and suppose* $C \geq 3$*. Fix any* $\mathbf{D} \in (\mathcal{S}^2_{++})^C$ *and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^\star) := \Big\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) \ : \ \mathcal{A}_{\mathbf{D}} \mathbf{z}^\star(x) + f\big(\mathbf{z}^\star(x)\big) = 0 \ \ \forall x \in \Omega \Big\}. \tag{10}$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^\star) \neq \emptyset$ *then it is an infinite-dimensional affine subset of* $C^1(\mathbb{R}^C; \mathbb{R}^C)$*. Specifically, for every* $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ *vanishing on* $\mathcal{R} := \mathbf{z}^\star(\Omega)$ *and every* $f \in \mathcal{F}$*, one has* $f + \varphi \in \mathcal{F}$*, and the space of such* φ *is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^\star(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^\star$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

Corollary 12 (What a stationarity fit can determine)**.** *Under the hypotheses of Theorem 11, the steady-state loss* (3) *constrains* f_θ *only on* $\mathcal{R}$*, a set of Lebesgue measure zero in* $\mathbb{R}^C$*. Any statement about* f_θ *off* $\mathcal{R}$*—in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 12 has a direct experimental consequence. The in-silico perturbation of Section 5.6 displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 11 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C-dimensional neighborhood of $\mathcal{R}$—dense temporal sampling, or interventional data.

F Linear stability and Turing analysis

Let $\mathbf{z}^\star \in \mathbb{R}^C$ be a spatially homogeneous equilibrium, so $f_\theta(\mathbf{z}^\star) = 0$, and write $\mathbf{J} := \nabla f_\theta(\mathbf{z}^\star)$. Linearising (5) about $\mathbf{z}^\star$ and expanding the perturbation in the basis (4) gives, for each $\mathbf{k}$, the finite-dimensional system

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\delta\hat{\mathbf{z}}(\mathbf{k}) \ = \ M(\mathbf{k})\,\delta\hat{\mathbf{z}}(\mathbf{k}), \qquad M(\mathbf{k}) \ := \ \mathbf{J} - \mathrm{diag}\big(q_1(\mathbf{k}), \ldots, q_C(\mathbf{k})\big), \tag{11}$$

with q_c as in (6). Define the growth rate $\mu(\mathbf{k}) := \max \mathrm{Re}\,\mathrm{spec}\, M(\mathbf{k})$ and its angular maximum $\mu^\star(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$; the latter is required because anisotropy makes q_c direction-dependent.

Definition 13 (Turing instability)**.** The operator exhibits a *diffusion-driven (Turing) instability* at $\mathbf{z}^\star$ if $\mu^\star(0) < 0$ and $\mu^\star(k) > 0$ for some $k > 0$.

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is $\lambda^\star = 2\pi/k^\star$ with $k^\star = \arg\max_k \mu^\star(k)$, converted to microns by the stored coordinate scale.

Proposition 14 (Pure diffusion cannot pattern)**.** *If* $f_\theta \equiv 0$ *then* $\mu^\star(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$*, strictly decreasing in* k*. No Turing instability exists.*

Proof. $M(\mathbf{k}) = -\mathrm{diag}(q_c(\mathbf{k}))$ is diagonal with real non-positive entries; its largest eigenvalue is $-\min_c q_c(\mathbf{k})$, and $q_c(\mathbf{k}) \geq \lambda_{\min}|\mathbf{k}|^2$ by Lemma 2. □

Proposition 15 (Two-channel criterion)**.** *Let $C = 2$ with isotropic tensors $\mathbf{D}_c = d_c I$, $\mathbf{J} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and write $\sigma = |\mathbf{k}|^2$. Then $\mu(\mathbf{k}) > 0$ for some $\sigma > 0$ while $\mu(0) < 0$ if and only if*

$$a + d < 0, \qquad ad - bc > 0, \qquad d_2 a + d_1 d > 2\sqrt{d_1 d_2 (ad - bc)}. \tag{12}$$

Proof sketch. M has trace $a + d - (d_1 + d_2)\sigma$ and determinant $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma + (ad - bc)$. The first two conditions give stability at $\sigma = 0$; the trace is then negative for all $\sigma > 0$, so instability requires $h(\sigma) < 0$ for some $\sigma > 0$. The minimum of the convex parabola h is attained at $\sigma^\star = (d_2 a + d_1 d)/(2 d_1 d_2)$ and is negative precisely under the third condition. □

Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Section 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreasing (Fig. 6c), and $\mu^\star(k) < 0$ for all $k > 0$ in every run.

Remark 16 (Why the negative result is expected). The training objective (3) rewards making the *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated as a fixed point rather than as the outcome of an instability, the fitted operator is driven toward dissipativity. Detecting pattern-forming instability would require observations away from steady state, consistent with Corollary 12.

G Discretization invariance of the encoder

The encoder must map measurements on an arbitrary point set to a field. Let μ be the sampling measure on Ω and consider the kernel-integral operator

$$(\mathcal{K}v)(x) \;=\; \int_\Omega \kappa_\theta(y - x) \odot v(y)\, \mathrm{d}\mu(y), \tag{13}$$

of which the graph layer is the empirical counterpart $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(x_j - x_i) \odot v(x_j)$.

Proposition 17 (Monte-Carlo consistency)**.** *If $\{x_j\}_{j=1}^N$ are i.i.d. draws from μ and $\kappa_\theta \odot v \in L^2(\mu)$, then for each x, $\mathbb{E}\big[(\mathcal{K}_N v)(x)\big] = (\mathcal{K}v)(x)$ and $\mathrm{Var}\big[(\mathcal{K}_N v)(x)\big] = O(N^{-1})$; consequently $\|\mathcal{K}_N v - \mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$.*

Proof. The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbiasedness and the $O(N^{-1})$ variance are immediate, and the norm bound follows by Chebyshev and Fubini. □

Proposition 17 is the sense in which the encoder is discretization-invariant: the population operator (13) depends on the geometry through $\kappa_\theta(y - x)$ alone and not on the sampling pattern, and the empirical operator converges to it as the sampling density increases, whatever the lattice. **This is an asymptotic statement about the encoder, and it is worth emphasising that it does not imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that it does not: $\mathbf{D}_\theta$ and f_θ are fitted to one tissue's stationary field, and Theorem 11 indicates they are unconstrained away from that field's range. Discretization invariance of the *architecture* and portability of the *fit* are distinct properties, and only the former is established here.

Proposition 18 (Splatting consistency)**.** *Let $w_\sigma(r) = e^{-r^2/2\sigma^2}$ and define the Nadaraya–Watson field $F_\sigma(u) = \sum_i w_\sigma(\|u - x_i\|) v_i / \sum_i w_\sigma(\|u - x_i\|)$. If $v_i = v(x_i)$ for $v \in C^2$ and the sampling density p is bounded away from zero, then $F_\sigma(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$.*

Proof sketch. Standard kernel-regression bias–variance decomposition in two dimensions: the bias term is $\frac{\sigma^2}{2}\left(\Delta v + 2\nabla v \cdot \nabla \log p\right) + o(\sigma^2)$ and the variance is $O((N\sigma^2)^{-1})$. □

The learnable bandwidth σ therefore trades a smoothing bias against sampling noise. Since the bias term is proportional to Δv, splatting contributes a further diffusive smoothing on top of the operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and one that the present work does not disentangle.

Complexity

Let N be the number of measured locations, G genes, C latent channels, $H \times W$ the lattice, E the number of graph edges and T the integration horizon. The costs per forward pass are: gene projection $O(NGd)$; graph layers $O(LEd)$; splatting $O(Nr^2C)$ for radius r; spectral blocks $O(BCHW \log HW)$; operator $O(TCHW \log HW)$; decoding $O(NCd + NGd)$. The operator term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used here (C=32, H=W=64, T=8) it is under 4% of the forward cost—consistent with the parameter count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin over the strongest baseline.

H Empirical details and full result tables

This appendix collects the complete result tables underlying Section 5. Tables are placed inline with `[h]` so that each appears with its heading.

H.1 Single-section model comparison

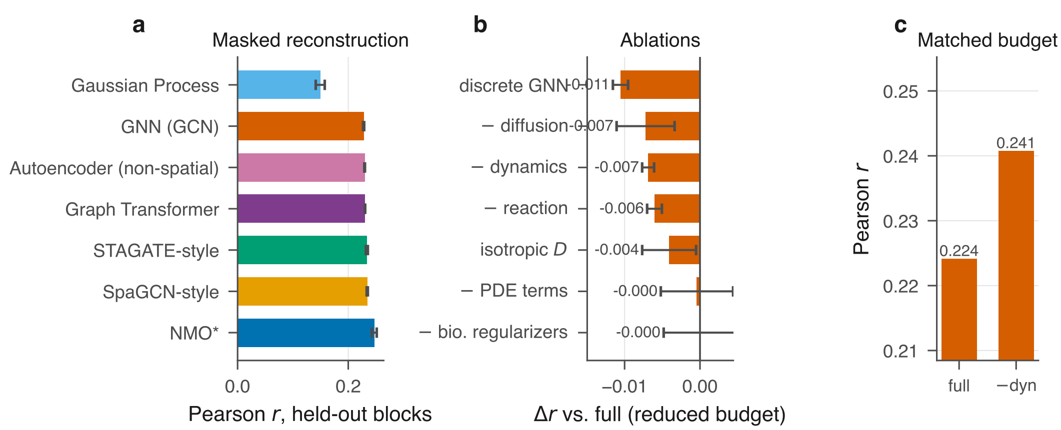

Figure 3: **Locating the effect** on the primary section, where all nine models are run. (a) Full model comparison. (b) Ablation deltas at the reduced budget. (c) The two decisive controls at the benchmark budget: disabling the operator (T=0) costs more than the full margin over the best baseline.

H.2 Numerical behavior

H.3 Paired statistics, numerics and biology

H.4 Dataset inventory

H.5 Ablation sweep

Full sweep at the reduced budget; the two matched-budget controls are reported in Section 6.

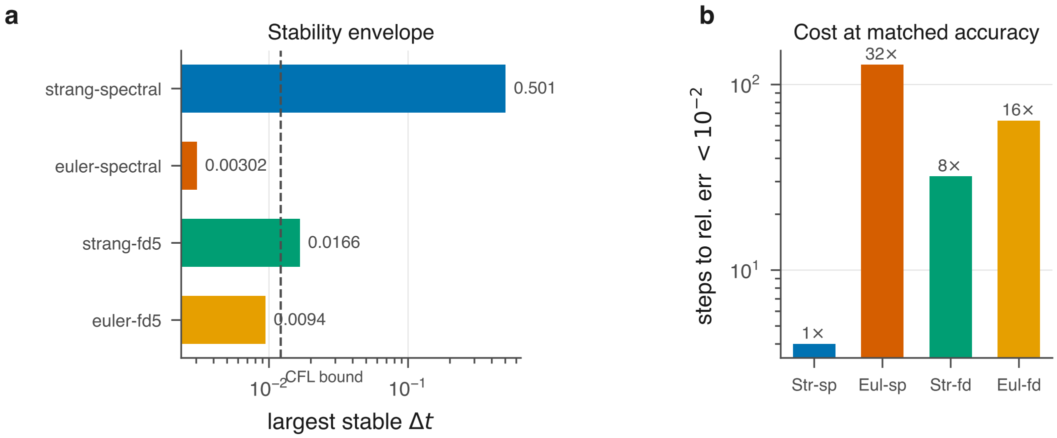

Figure 4: **What the exact spectral solve buys.** (a) Largest stable step size per scheme against the CFL bound for the explicit five-point Laplacian. (b) Steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme.

Table 3: Paired comparison of NMO against each baseline, with the tissue section as the unit of analysis. Δ is the mean paired difference in Pearson r (positive favors NMO) with a percentile bootstrap 95% CI over sections; d_z is Cohen's effect size for paired designs; p is the Wilcoxon signed-rank test and p_{Holm} its Holm–Bonferroni correction across the baseline family.

baseline	n	Δr	95% CI	d_z	p	p_{Holm}	wins
Neural field (SIREN)	17	+0.0622	[+0.0481, +0.0772]	1.97	1.5e-05	7.6e-05	17/17
GP, multi-bandwidth	17	+0.0453	[+0.0371, +0.0538]	2.49	1.5e-05	7.6e-05	17/17
Autoencoder (non-spatial)	17	+0.0159	[+0.0085, +0.0237]	0.96	0.0017	0.0034	13/17
GNN (GCN)	17	+0.0155	[+0.0093, +0.0220]	1.13	0.00011	0.00032	16/17
STAGATE-style	17	+0.0146	[+0.0071, +0.0225]	0.88	0.0032	0.0034	13/17

Table 4: Numerical behavior of the integrator variants. The largest stable step is measured by sweeping Δt over $[10^{-4}, 0.501]$ and testing boundedness over 300 steps; the CFL reference is $h^2/(4\lambda_{\max}) = 0.01215$ for the explicit five-point Laplacian. **strang-spectral never destabilized within the sweep** ($\Delta t \leq 0.501$), so its entry is a lower bound set by the grid rather than a measured threshold, consistent with the unconditional stability of Proposition 5. Cost is the number of steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme; the achieved error is reported alongside, since the step grid is coarse and the schemes do not land on the same accuracy. The empirical limit for `euler-fd5` tracks the CFL bound, confirming the analysis.

scheme	largest stable Δt	vs. CFL	steps to 10^{-2}	achieved err.	cost
strang-spectral	≥0.501	≥41×	4	8.4e-03	1×
euler-spectral	0.00302	0.25×	128	2.3e-03	32×
strang-fd5	0.0166	1.4×	32	9.1e-04	8×
euler-fd5	0.0094	0.77×	64	6.1e-03	16×

H.6 Cross-tissue and cross-resolution transfer

H.7 Developmental forecasting

Horizon sweep for E9.5 → E10.5. Consecutive MOSTA stages are distinct embryos, so the comparison is at the level of the rasterized field after isotropic normalization to a common frame.

Table 5: Biological preservation of the reconstruction, averaged over 4 sections with reference annotations (Space Ranger clusters, Xenium clusters, and curated anatomical regions). Predicted expression at held-out locations is clustered and scored against the reference; ARI retention is the ratio to the score obtained from the measured data, so 1.0 would mean the reconstruction is as informative as the measurement. Marker AUROC asks whether the predicted field still discriminates a region using that region's measured markers; k-NN preservation is the overlap of neighborhoods in measured versus predicted expression space; $|\Delta C|$ is the error in Geary's C.

model	ARI ↑	ARI ret. ↑	NMI ↑	marker AUROC ↑	k-NN pres. ↑	$\lvert\Delta C\rvert$ ↓
GNN (GCN)	0.183 $\pm$ 0.065	0.495 $\pm$ 0.318	0.365 $\pm$ 0.071	0.764 $\pm$ 0.028	0.181 $\pm$ 0.052	0.727 $\pm$ 0.041
GP, multi-bandwidth	0.183 $\pm$ 0.045	0.408 $\pm$ 0.145	0.395 $\pm$ 0.041	0.775 $\pm$ 0.086	0.221 $\pm$ 0.059	0.727 $\pm$ 0.038
STAGATE-style	0.179 $\pm$ 0.041	0.467 $\pm$ 0.224	0.375 $\pm$ 0.058	0.768 $\pm$ 0.026	0.184 $\pm$ 0.053	0.715 $\pm$ 0.036
Neural field (SIREN)	0.091 $\pm$ 0.060	0.182 $\pm$ 0.110	0.219 $\pm$ 0.131	0.696 $\pm$ 0.058	0.106 $\pm$ 0.027	**0.537** $\pm$ **0.062**
NMO (ours)	**0.197** $\pm$ **0.051**	**0.518** $\pm$ **0.276**	**0.407** $\pm$ **0.086**	**0.779** $\pm$ **0.027**	**0.228** $\pm$ **0.082**	0.771 $\pm$ 0.051

Table 6: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_01	MERFISH, 500-gene pane	Mus	23,376	500	4178×4097	single cell
merfish_allen_05	MERFISH, 500-gene pane	Mus	45,896	501	6127×5166	single cell
merfish_allen_10	MERFISH, 500-gene pane	Mus	42,070	503	8402×5441	single cell
merfish_allen_14	MERFISH, 500-gene pane	Mus	94,357	500	9048×5717	single cell
merfish_allen_18	MERFISH, 500-gene pane	Mus	44,309	500	7980×4825	single cell
merfish_allen_26	MERFISH, 500-gene pane	Mus	79,282	501	9460×5685	single cell
merfish_allen_30	MERFISH, 500-gene pane	Mus	91,017	501	9870×5843	single cell
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837×6094	single cell
merfish_allen_36	MERFISH, 500-gene pane	Mus	115,675	503	10074×6559	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000×6961	single cell
merfish_allen_42	MERFISH, 500-gene pane	Mus	75,433	503	9875×7086	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806×6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500×4950	bin50 (25 um)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975×2650	bin50 (25 um)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_human_breast	Visium (55 um spots, h	Homo	4,869	2,087	6395×6750	spot (multi-cell)
visium_mouse_brain	Visium (55 um spots, 1	Mus	2,691	2,079	4822×6379	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212×3128	single cell

Table 7: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The − *dynamics* row is the critical control: it disables the operator entirely (T=0) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than the main benchmark** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson r ↑	Δr
Full model	2.20M	**0.233** $\pm$ **0.005**	–
− dynamics (T=0)	2.20M	0.227 $\pm$ 0.001	-0.007
− diffusion term	2.20M	0.226 $\pm$ 0.004	-0.007
− reaction term	2.19M	0.227 $\pm$ 0.001	-0.006
− PDE constraints	2.20M	0.233 $\pm$ 0.005	-0.000
− biological regularizers	2.20M	0.233 $\pm$ 0.005	-0.000
isotropic D	2.20M	0.229 $\pm$ 0.004	-0.004
state-dependent D	2.20M	0.236 $\pm$ 0.004	+0.003
discrete GNN operator	0.61M	0.223 $\pm$ 0.001	-0.011
latent C=8	1.07M	0.213 $\pm$ 0.005	-0.021
latent C=16	1.30M	0.220 $\pm$ 0.002	-0.013
latent C=64	5.77M	0.237 $\pm$ 0.000	+0.004

Table 8: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
source (in-domain)			
STAGATE-style	0.212 ± 0.001	0.249 ± 0.000	0.358 ± 0.001
NMO (ours)	0.209 ± 0.004	0.251 ± 0.003	0.365 ± 0.004
target (zero-shot)			
STAGATE-style	0.093 ± 0.001	0.397 ± 0.002	0.266 ± 0.002
NMO (ours)	0.033 ± 0.012	0.409 ± 0.005	0.219 ± 0.017
target (decoder-only fine-tune)			
STAGATE-style	0.117 ± 0.007	0.393 ± 0.000	0.365 ± 0.003
NMO (ours)	0.041 ± 0.019	0.406 ± 0.004	0.282 ± 0.017
target (in-domain oracle)			
STAGATE-style	0.147 ± 0.001	0.375 ± 0.000	0.427 ± 0.001
NMO (ours)	0.136 ± 0.001	0.397 ± 0.002	0.329 ± 0.003
target (training-mean floor)			
mean	-0.000 ± 0.000	0.404 ± 0.002	0.254 ± 0.001

Table 9: Cross-resolution transfer: 55 μm Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
source (in-domain)			
NMO (ours)	0.256 ± 0.009	0.246 ± 0.002	0.384 ± 0.007
STAGATE-style	0.242 ± 0.001	0.248 ± 0.001	0.364 ± 0.001
target (zero-shot)			
STAGATE-style	0.282 ± 0.000	1.765 ± 0.001	0.208 ± 0.001
NMO (ours)	0.139 ± 0.015	1.863 ± 0.026	0.125 ± 0.004
target (in-domain oracle)			
NMO (ours)	0.259 ± 0.006	1.752 ± 0.007	0.556 ± 0.006
STAGATE-style	0.256 ± 0.001	1.760 ± 0.002	0.569 ± 0.001
target (training-mean floor)			
mean	0.000 ± 0.000	1.872 ± 0.000	0.101 ± 0.001

Table 10: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterized field after isotropic normalization to a common frame; it inherits the error of that coarse registration. *Persistence* predicts E10.5 = E9.5 and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NMO	T=0	0.082 ± 0.001	0.320	74%
NMO	T=2	0.089 ± 0.000	0.305	72%
NMO	T=4	0.092 ± 0.000	0.297	73%
NMO	T=8	0.094 ± 0.000	0.289	72%
NMO	T=16	0.095 ± 0.001	0.283	72%
NMO	T=32	0.090 ± 0.003	0.285	70%

H.8 In-silico morphogen source

Split-half protocol of Section 5.6: the perturbation direction is defined from a random half of each pathway and enrichment scored on the held-out half, against a permutation null over matched gene sets.

Table 11: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway's genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $^*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 ± 6.0	49.7	0.475	575 ± 154
FGF	19	72.6 ± 0.8	50.3	0.992	618 ± 58
SHH	15	56.6 ± 6.7	50.4	0.698	466
WNT	26	56.4 ± 3.9	49.9	0.762	649 ± 143

H.9 Perturb-seq consistency

An out-of-context probe: Norman et al. profiled non-spatial K562 cells whereas the operator is fitted to tissue. With $n = 7$ testable perturbations the comparison is underpowered and we report it as inconclusive.

Table 12: Agreement between the model's counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe**: the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signaling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NMO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

H.10 Latent fields

I Reproducibility

All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

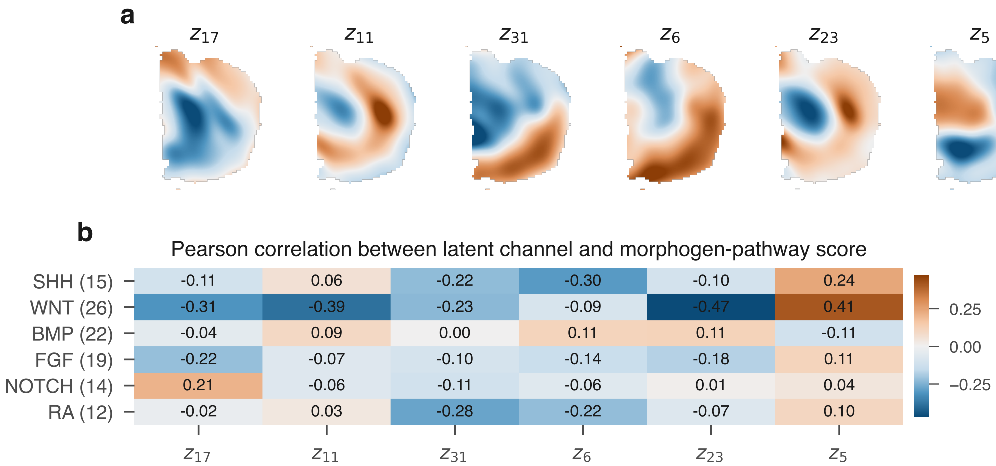

Figure 5: Latent channels of the relaxed field (a) and their correlation with canonical morphogen-pathway scores (b). Section 5.6 shows these correlations do not survive a causal split-half test.

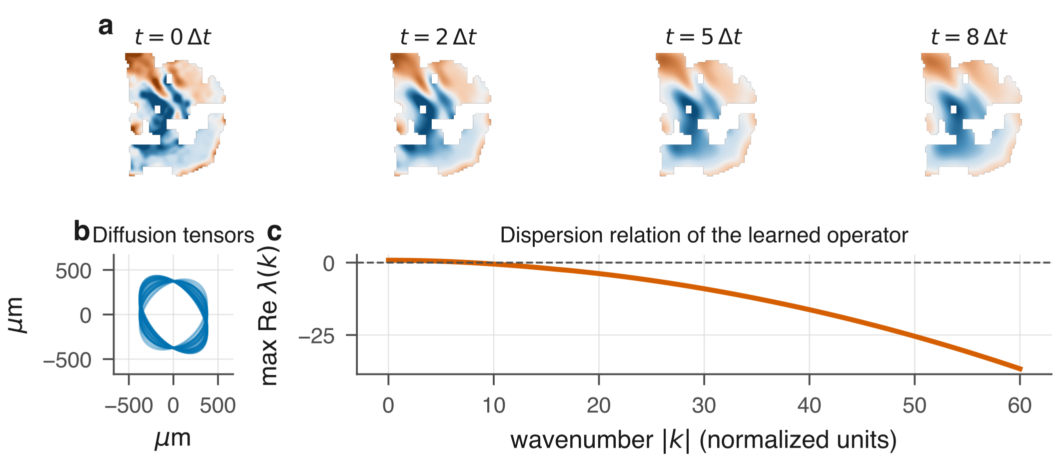

Figure 6: **The fitted operator.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation of the linearized operator, which decreases monotonically in $|\mathbf{k}|$ and so supports no Turing band.